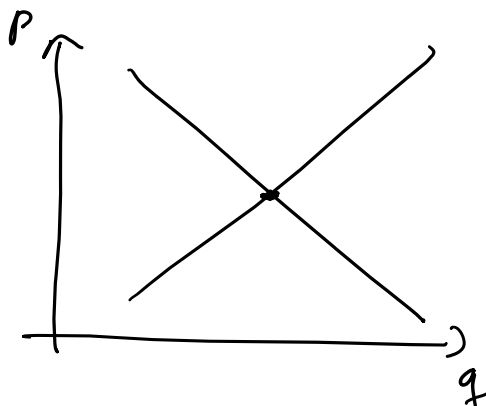
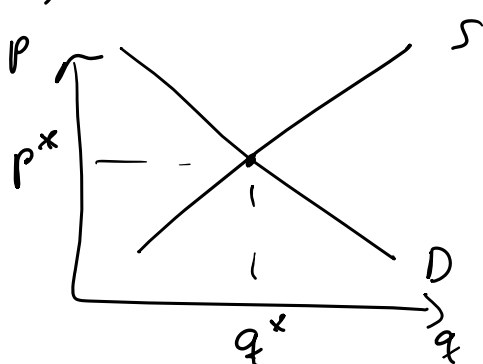


What is Macroeconomics?

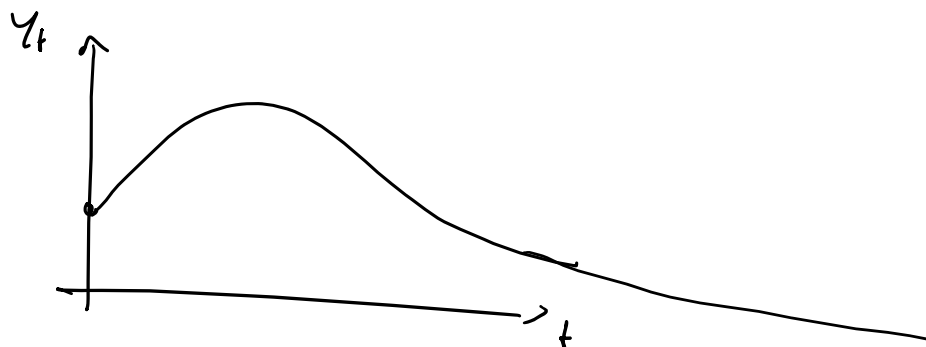
- Macro is about aggregates.

$$\underline{C_t} = \sum_i C_{it}$$
$$\underline{Y_t} = \sum_i y_{it}$$

- Macro is often studied in general equilibrium (GE)

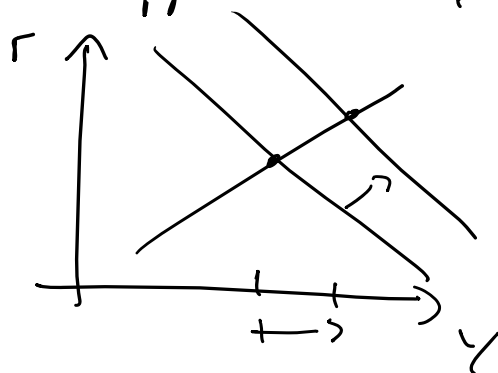


- Macro is often studied in dynamic environments



- Macro is quantitative
(as opposed to qualitative)

(as opposed to qualitative)



⊖ Business Cycles
 ↳ Monetary Policy
 → Fiscal Policy

⊖ Economic Growth.

- Macro labor
- Structural change
- inequality
- climate change
- . . .

Adjacent Fields

- International Trade
 " Finance
- Macro Finance
- Macro-Development

Brief History of Macroeconomics

Brief History of Macroeconomics (mostly business cycle macro)

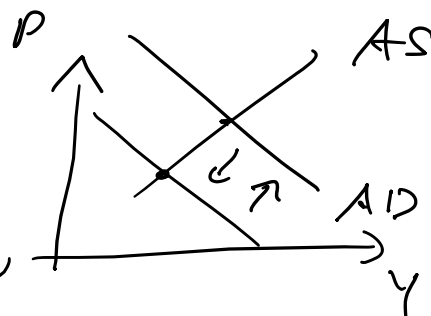
Prior to the 1930s (Classical Economics)

- Adam Smith, Alfred Marshall
- no macro measurement (GDP had not been invented)

Great Depression 1929

② 1930s Keynesian Revolution

- General Theory
- Aggregate Demand, Multipliers, Animal Spirits
- Advocates for countercyclical Fiscal Policy



Measurement Revolution

- Development of the National Income and Product (NIPA) accounts

- Input-Output Tables

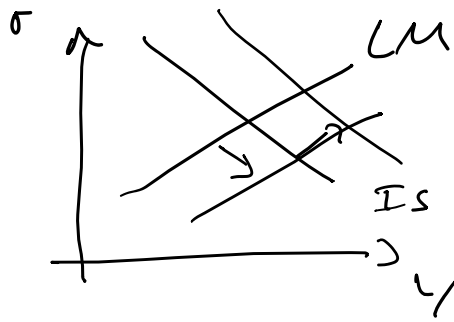
Product → Sectors



□ □

Σ

John Hicks: "Mr. Keynes and the Classics:
A suggested Interpretation"

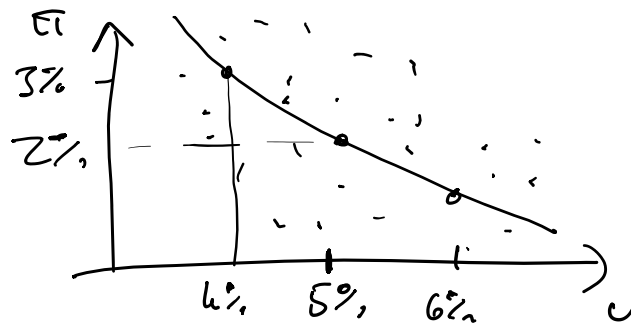


1940s-1960s Keynesian Consensus

- IS-LM Model (r, Y) "fine-tuning" economy
- Okun's law

$$\frac{\Delta \ln Y}{\ln Y} \approx -2\% \rightarrow \text{u-rate} \uparrow 1\%$$

- Phillips Curve (PC)



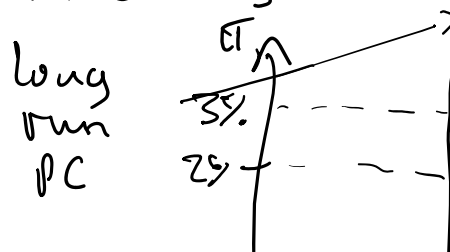
"Menu" of Policy Options

$$\boxed{r, Y, u, \pi}$$

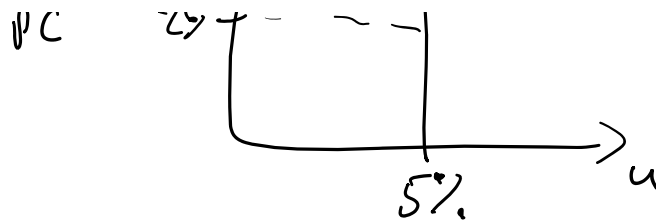
1960s-1970s Rise of Monetarism

- Milton Friedman (U Chicago)
- Edmund Phelps

In the long run the ~~the~~ PC is vertical

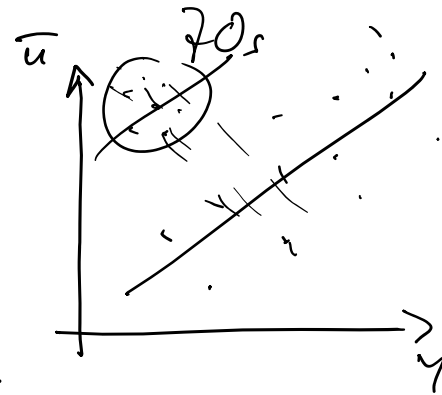
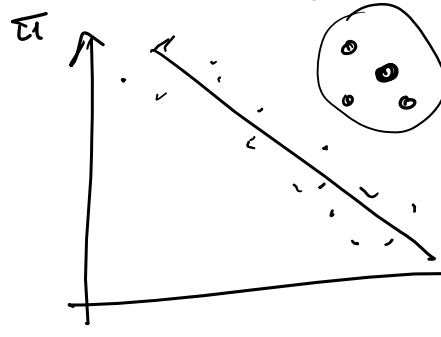


"Inflation is always and everywhere a monetary phenomenon"



everywhere
a monetary
phenomenon u

- in 1970s Oil Shocks



- 1970s - 1990s (New Classical Economics and Rational Expectations)

Robert Lucas

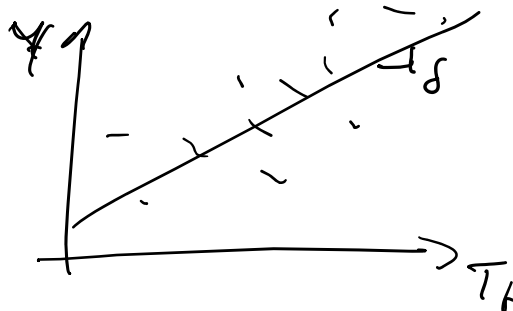
- the Lucas Critique

$$Y_t = \alpha + \beta Y_{t-1} + \gamma \pi_{t-1} + \delta T_t + \varepsilon_t$$

$$\hat{Y}_t = \alpha + \beta \hat{Y}_{t-1} + \gamma \hat{\pi}_{t-1} + \delta \hat{T}_t$$

using ad hoc econometric models
are going to produce conceptually
incorrect counterfactuals

$$Y_t = \alpha + \delta T_t + \varepsilon_t$$



$$T_t = \tau \cdot Y_t$$

- Rational Expectations

– Rational Expectations

$$E[y] = \int y' g(y) dy$$

– “Time to build and aggregate fluctuations”

Woodford & Prescott (1982)

RBC paradigm (Real business cycles)

Prescott + Summers debate

Schism → Saltwater School (Summers)

→ Freshwater School (Chicago, Minnesota, Carnegie Mellon)

1990s – today

New Keynesian model

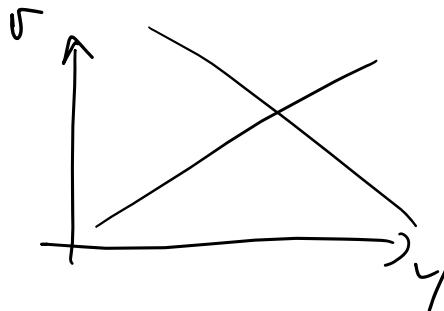
↳ Galí, Woodford

RBC – capital

+ sticky prices

→ monetary nonneutrality

⇒ Monetary Policy affects real variables



Empirical Macroeconomics $\left(\frac{y_t}{n_t} \right)$

Empirical Macroeconomics

VARs $\leftarrow$

$$Y_t = \begin{pmatrix} y_t \\ u_t \\ r_t \end{pmatrix}$$

Vector Autoregressions

Sims, Sargent

Microeconomic methods in macro

$$Y_t = \alpha + \beta Y_{t-1} + u_t$$

Macro branching out into other fields

↳ climate

↳ trade

↳ geoeconomics

